Replication Report: (title not recorded)

OSTI 2475938 | Coverage 6/10, Agreement 7/10

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1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 2475938

2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `2475938-Updated-Virophage-Taxonomy-and-Distinction-from/replication_plan.pdf`.

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper's computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/tie`

Paper-directory README excerpt

Updated Virophage Taxonomy and Distinction from Polinton-like Viruses

- **OSTI ID:** 2475938
- **Paper:** Roux et al., *Biomolecules* 13:204 (2023)
- **Rank:** #9
- **Original replication-score target:** 9/10
- **Realised score (this run):** ~6/10 (simplified reference set)

Why This Paper

This genomics paper uses publicly available genome sequences, computational analysis, and quantitative taxonomic criteria, allowing for fully automated AI replication and validation.

Replication Plan (high level)

Collect genome datasets → predict ORFs (Prodigal) → apply HMM profiles for marker genes (MCP, ATPase, PRO, penton + PLV HMM) → align & trim markers → build ML phylogenies (IQ-TREE) → confirm virophage monophyly and virophage/PLV discrimination.

Status

- [x] Paper reviewed
- [x] Data identified (22 NCBI GenBank accessions; 13 retained after size filtering)
- [x] Code implemented (Prodigal + HMMER + MAFFT + trimAl + IQ-TREE)
- [x] Results validated (congruent 4-marker phylogeny; clean HMM separation of virophages vs PLV/adintovirus)

This Replication

Run on **CherryRd** (macOS, conda env ‘virophage‘), total wall time < 20 min compute (+ ~5 min download). Deliverables:

- **Report PDF:** [‘replication/report/report.pdf‘](replication/report/report.pdf) (6 pages)
- **Figures:** ‘replication/report/phylogeny_4markers.pdf‘, ‘replication/report/hmm_heatmap.pdf‘
- **Classification table:** ‘replication/results/hmm_summary.tsv‘
- **Trees (Newick):** ‘replication/results/trees/{MCP,ATPase,PRO,Penton}.treefile‘
- **Alignments:** ‘replication/results/markers/{MCP,ATPase,PRO,Penton}.{aln,trim}‘

Key findings (replicated)

1. **HMM classification matches the paper.** 7/13 genomes (Sputnik, Mavirus, SW01, YSLV1–4) hit all four virophage marker HMMs strongly (MCP   472 bits); the remaining 6 (TVSG_01 PLV, three *Chrysochromulina parva* virophages, adintovirus, and two partial/mis-labelled GenBank records) have **zero** hits above inclusion to any of the 19 virophage HMMs or the PLV HMM. 2. **Virophage–PLV distinction is reproducible.** Virophages score   82–765 bits on the four morphogenesis markers; PLVs / adintoviruses score 0 — directly confirming the paper's statement that the marker HMMs separate the two classes. 3. **Four-marker phylogenetic congruence.** MCP, ATPase, PRO, and penton

[\textit{README truncated}]

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper's resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Not recorded.

5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper's 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Not recorded.

7 Score and friction-point tagging

- **Coverage:** 6/10 — Not recorded.
- **Agreement:** 7/10 — Not recorded.

Friction points encountered (taxonomy tags):

- **F5**: Force-field / hyperparameter drift (unspecified configuration)

8 Follow-on questions

- Not recorded.

9 Reproducibility pointers

- **Paper directory:** `2475938-Updated-Virophage-Taxonomy-and-Distinction-from`
- **Replication artefacts:**
  - `/Users/stevens/Dropbox/REPLICATE-PROJECT/2475938-Updated-Virophage-Taxonomy-and-Disti`
- **Replication-dir hint from JSONL:** `/Users/stevens/Dropbox/REPLICATE-PROJECT/2475938-Updated-V`
- **Status:** tier_lift_v2_2026_04_27

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.